

EMMANUEL ADELOJU

eadeloju[[@](#)]asu[[dot](#)]edu | [Website](#) | [LinkedIn](#) |

Human-AI Interaction | UX Research | LLM Evaluation

EDUCATION

PhD, Learning Literacies and Technology Arizona State University	Expected 2027
MSc, Cell Biology and Genetics University of Lagos	2021 - 2022
Postgraduate Diploma in Education National Teachers Institute Kaduna	2018 - 2019
BSc, Biochemistry Osun State University	2012 - 2016

SKILLS

- User Research:** Experimental & study design, controlled studies, randomized group assignment, hypothesis testing, A/B testing, surveys, structured interviews, focus group, contextual inquiries, usability evaluation, mixed-methods research
- Quantitative Analysis & Statistics:** Inferential & descriptive statistics, comparative group analysis, dose/dependent-variable response analysis, data cleaning & interpretation
- AI:** Machine Learning, LLM Evals
- Software:** MAXQDA, Qualtrics, Figma, Miro, Excel
- Programming:** Python, R, SPSS

SELECTED RESEARCH & TEACHING EXPERIENCE

DISSERTATION RESEARCH	09/2025 - Present
- Led a longitudinal qualitative study (6 months) of how educators adopt and make sense of LLM-based tools, conducting mixed-methods UX research (interaction analysis, semi-structured interviews, document analysis) and contextual inquiries across a multi-phase professional-learning journey from first exposure to real-world implementation. - Designed and ran a comparative multiple-case study tracing users across three adoption stages, <i>learner</i>, <i>designer</i>, <i>implementer</i>, to map how mental models, workflows, and trust in AI evolve over time; surfaced stage-specific usability barriers and behavioral shifts in human-LLM interaction. - Captured and analyzed screen recordings and think-aloud transcripts of user-LLM sessions, developing a parsimonious, reliability-tested behavioral codebook (6 codes) to identify friction points, breakpoints, and pivotal moments in real-time AI tool use. - Identifying recurring interaction breakdowns (e.g., tool failures, output-trust gaps, prompt-iteration struggles) and traced users' adaptive recovery strategies, producing evidence-based insights into how non-expert users troubleshoot and re-engage with generative-AI tools. - Translating qualitative findings into an operationalized analytic framework linking observable user behaviors to higher-level constructs, creating an auditable observation-to-insight pipeline suitable for cross-participant comparison and stakeholder reporting.	

- Synthesizing findings on data-sensemaking workflows with LLMs into actionable design implications for AI tool onboarding, scaffolding, and trust calibration for first-time and recurring users.
- Designing SciGroundBench, a structured LLM evaluation benchmark operationalizing data-sensemaking into a four-dimensional rubric across six science.

UL RESEARCH INSTITUTES & ARIZONA STATE UNIVERSITY

Program & Research Lead Analyst: Xplorlabs Fellowship

2024 – 2025

- Spearheaded mixed-methods UX research and end-to-end program evaluation for the Xplorlabs Fellowship, a multi-stakeholder K-12 safety science initiative, engaging 23 educators and 5 student fellows across 8 US states to generate evidence-based insights for ASU and ULRI leadership.
- Executed quantitative analysis of formative and summative survey data (descriptive and inferential statistics) to assess teacher engagement, implementation challenges, and pedagogical alignment; translated findings into prioritized recommendations adopted by program leads.
- Co-developed and validated survey instruments against evaluation objectives across measurement waves, ensuring construct validity, instrument coherence, and longitudinal comparability between formative and summative assessments.
- Conducted in-depth interviews with student fellows and synthesized findings into journey maps spanning project ideation, pivots, community engagement, and implementation, applying thematic coding grounded in the Action-Oriented Pedagogies (AOP) framework to surface design insights on agency, partnerships, and barriers to youth-led action.
- Field documentation protocols (photo capture, observation notes, artifact logging) across workshops, convenings, conferences, producing a longitudinal evidence corpus that enabled triangulation across quantitative and qualitative streams.
- First-authored comprehensive evaluation reports synthesizing program impact, teacher engagement, and curricular alignment for cross-functional stakeholders (university leadership, industry research partners, K-12 educators); presented insights at conferences.
- Facilitated a research-grounded professional development session at Community of Practice co-design workshop, translating ongoing evaluation insights into actionable implementation guidance for educators adopting AOP in science classrooms.

GRADUATE TEACHING ASSISTANT

Spring 2025

COE 502: Introduction to Data Analysis (Dr. Audrey Beardsley)

- Led weekly applied statistics lab sections for graduate students across an 18-module quantitative curriculum spanning descriptive statistics, probability distributions, hypothesis testing, correlation, linear regression, one-way & factorial ANOVA, chi-square, and nonparametric methods and delivered end-to-end workflows in SPSS and Excel from data wrangling through interpretation and write-up.
- Taught full modules independently, translating statistical theory into decision frameworks for selecting appropriate quantitative methods, drawing valid inferences from noisy real-world data.
- Tutored students on reliability, validity, and threats to validity in measurement design, reinforcing the instrumentation that underpins UX survey construction, scale validation, and quantitative usability metrics.

SELECTED COURSEWORK

APM 527: *Deep Neural Network*

EE 549: *Statistical Machine Learning – Theory to Practice*

PAF 513: *Foundations of Data Science (in R programming)*

COE 502: *Introduction to Data Analysis*

STP 530: *Applied Regression Analysis*

SELECTED AWARDS

CVPR26 Broadening Participation Scholarship Award

2026 College Dissertation Research Grant

2025 Mary Lou Fulton Endowment Scholarship

2025 Graduate Support Award

INVITED GUEST TALK

2026 AI for Social Science Research, Arizona State University

2025 ASU Preparatory Academy: Introduction to LLM for Datasensemaking

PUBLICATION & CONFERENCE

On website